MECH 191 — Top 5 Board Terms by Week

The five most critical vocabulary terms to emphasize each week

MECH 191 — Metallurgy

Leeward Community College · Math and Science Division · Mechatronics Program

For instructor use — board reference, not a student handout

How to use this list

One page per week — just the five vocabulary terms most worth writing on the board that day, picked as the anchor concepts students need to walk away with.

Week 1 — Atomic Structure & Bonding

Alloy	An intentional mixture of two or more elements engineered for desired properties.
Metallic bond	Electrons form a shared 'electron sea' — explains conductivity and ductility.
Grain / grain boundary	A single crystal domain; the boundary between differently oriented grains.
Body-centered cubic (BCC)	A crystal structure with atoms at the eight corners of a cube plus one atom in the center; moderately packed (~68%), found in ferrite (α-iron), chromium, and tungsten — generally more brittle at low temperature than FCC metals.
Face-centered cubic (FCC)	A crystal structure with atoms at the eight corners of a cube plus one atom centered on each of the six faces; densely packed (~74%), found in austenite (γ-iron), aluminum, copper, and nickel — generally the most ductile common structure.

Week 2 — Mechanical Properties & Behavior

Strength	A material's ability to resist an applied load without yielding or fracturing; most commonly reported as yield strength (the stress at which permanent deformation first begins) or ultimate tensile strength (the maximum stress the material reaches before necking).
Hardness	A material's resistance to localized permanent (plastic) deformation at its surface, measured by pressing a standardized indenter into the material under a known load and reading the size or depth of the resulting impression (Rockwell, Brinell, Vickers).
Ductility	A material's ability to undergo large plastic deformation — especially stretching — before it fractures, reported from a tensile test as % elongation or % reduction in area; high-ductility metals can be drawn into wire or deep-drawn into sheet parts.
Toughness	A material's ability to absorb energy and deform plastically before fracturing, combining strength and ductility; measured as the area under a material's stress-strain curve, or directly with an impact test (Charpy/Izod).
Brittleness	A material's tendency to fracture with little or no plastic deformation once its strength is exceeded — the opposite of ductility and toughness; brittle fractures happen suddenly and leave a flat, crystalline surface rather than necking or dimpling.

Week 4 — Carbon & Alloy Steels

Ferrite / austenite (BCC/FCC iron)	Low-temp BCC iron phase vs. high-temp FCC iron phase — the basis of steel allotropy.
Carbon equivalent (CE)	CE = %C + %Mn/6 + (%Cr+%Mo+%V)/5 + (%Ni+%Cu)/15 — determines weldability/preheat needs.
Martensite	A hard, brittle, supersaturated body-centered tetragonal microstructure that forms when austenite is cooled too fast for diffusion to occur, trapping carbon in the iron lattice; requires tempering to restore usable toughness.
Hardenability	How deeply a steel can be hardened through its cross-section (different from hardness).
Quenching	Rapid cool from austenite — maximizes hardness (forms martensite).

Week 5 — Stainless Steels & Cast Irons

Passive chromium oxide layer (Cr_2O_3)	Self-repairing surface film that makes stainless steel corrosion-resistant; requires ≥10.5% Cr.
Austenitic stainless (300 series)	FCC, non-magnetic, not hardenable by heat treatment — e.g. 304, 316.
Sensitization	Chromium carbide precipitation at grain boundaries during welding — depletes nearby Cr, causing intergranular corrosion susceptibility.
Gray cast iron	Carbon as graphite flakes — good damping/machinability, low tensile strength.
Ductile (nodular) iron	Mg/Ce addition makes graphite form as spheroids — much tougher than gray iron.

Week 6 — Nonferrous Metals & Phase Diagrams

Aluminum alloy series (1xxx–7xxx)	Designation system by principal alloying element; heat-treatable vs. work-hardened series.
T6 temper	Solution treat + age — most common heat-treated condition for 6061/7075 aluminum.
Iron-carbon phase diagram	Maps stable phases across carbon content and temperature — ferrite, austenite, cementite, pearlite.
Eutectoid point	0.77% C at 727°C — austenite transforms simultaneously to ferrite + cementite.
Brass (Cu-Zn) vs. Bronze (Cu-Sn / Cu-Al)	The two major copper alloy families — know composition and typical uses.

Week 7 — Mechanical Testing Methods

Rockwell hardness test	Presses a diamond cone or hardened ball indenter into the surface under a minor then major load and reads hardness directly off the depth of penetration; fast and simple, the standard shop-floor check for finished, heat-treated steel parts.
Tensile test	A mechanical test in which a specimen is pulled along its axis at a controlled rate until it necks and fractures, while load and elongation are continuously recorded to determine yield strength, ultimate tensile strength, elastic modulus, and ductility.
S-N curve (Wöhler curve)	Plots stress amplitude vs. cycles to failure — used to find the fatigue limit.
Charpy impact test	An impact toughness test in which a notched specimen is supported horizontally at both ends and struck from behind the notch by a swinging pendulum, measuring the energy absorbed in fracture.
Ductile-brittle transition temperature (DBTT)	Temperature below which a material's fracture behavior shifts from tough to brittle.

Week 8 — Non-Destructive Testing (NDT) Methods

Visual testing (VT)	The foundational inspection method — naked eye, lighting, magnification, borescopes.
Liquid/dye penetrant testing (PT)	Capillary action draws penetrant into surface-breaking defects; color contrast vs. fluorescent.
Magnetic particle testing (MT)	Detects surface/near-surface defects in ferromagnetic materials via flux leakage; yoke vs. prod methods.
Ultrasonic testing (UT)	Sound waves detect internal defects; A-scan, phased array (PAUT), straight vs. angle beam.
Radiographic testing (RT)	X-ray/gamma imaging of volumetric defects; film vs. digital radiography.

Week 9 — Quality Standards & Inspection

Indication → Discontinuity → Defect → Reject	The precise evaluation sequence — memorize the distinct legal meaning of each term.
Porosity	A weld, casting, or PM defect consisting of small gas pockets (voids) trapped in the solidified or sintered metal, usually caused by dissolved or entrapped gas that could not escape before solidification finished.
Lack of fusion (LOF)	A weld defect in which the weld metal fails to bond completely with the base metal or with a previously deposited pass, leaving an unfused interface, usually caused by insufficient heat input, poor technique, or surface contamination.
Non-Conformance Report (NCR)	Formal documentation raised when a component is rejected.
Traceability	Linking an inspection record to the physical component via unique identifiers.

Week 10 — Casting Processes & Defect Recognition

Sand casting	Pattern, mold, cores, gating system — flexible, low tooling cost, rough finish.
Die casting (hot chamber vs. cold chamber)	High-pressure injection into a reusable die — limited to lower-melting alloys.
Investment (lost-wax) casting	Ceramic shell over a wax pattern — best surface finish and accuracy of any casting process.
Gating system	The complete network of channels in a sand-casting mold that carries molten metal from the pouring cup into the mold cavity.
Shrinkage porosity / gas porosity	Two major casting defect types — irregular voids vs. smooth spherical voids.

Week 11 — Metal Forming Processes

Hot working vs. cold working	No work hardening vs. work hardening — know the trade-offs of each.
Rolling	Reducing a metal's thickness (or forming its cross-section) by passing it between rotating rolls; performed hot or cold and used to produce plate, sheet, and structural shapes.
Forging	A compressive shaping process that hammers or presses metal into shape, producing the highest-integrity wrought parts; performed with open or closed dies.
Extrusion (direct vs. indirect)	Forcing metal through a die to produce constant-cross-section profiles.
Drawing	Pulling metal through a die (or forming it with a punch) under tension to reduce its cross-section or reshape it — as in wire drawing, tube drawing, and deep drawing.

Week 12 — Machining

Turning	A machining process in which a cutting tool removes material from a rotating workpiece on a lathe to produce cylindrical or conical external and internal surfaces.
Milling	A machining process in which a rotating multi-tooth cutter removes material from a stationary or moving workpiece to produce flat surfaces, slots, pockets, and contoured features.
Cutting speed (Vc)	The speed at which the cutting edge moves relative to the workpiece surface, usually in meters per minute (m/min); set primarily by the tool material and the workpiece being cut.
Feed (f)	The distance the cutting tool advances relative to the workpiece per revolution (turning) or per tooth (milling), which — together with the tool's nose radius — largely determines the resulting surface finish.
Surface roughness: Ra and Rz	Arithmetic mean roughness vs. peak-to-valley height — the two key surface finish parameters.

Week 13 — Powder Metallurgy

Sintering	Densification by solid-state atomic diffusion below the melting point — not melting.
Green compact / green strength	The fragile, pressed-but-unsintered powder part before sintering.
Hot isostatic pressing (HIP)	Simultaneous heat + isostatic gas pressure — eliminates porosity, approaches wrought properties.
Additive manufacturing	A family of layer-by-layer metal powder or wire fusion processes that build a part directly from a 3D model, with no tooling required.
Water atomization vs. gas atomization	Powder production methods producing irregular vs. spherical particle shapes.

Week 14 — Welding Processes

SMAW (Shielded Metal Arc Welding / Stick)	Coated consumable electrode; flux forms shielding gas and slag.
GMAW (Gas Metal Arc Welding / MIG)	Continuous bare wire with shielding gas; transfer modes: short circuit, spray, pulsed.
GTAW (Gas Tungsten Arc Welding / TIG)	Non-consumable tungsten electrode; highest quality; AC for aluminum, DCEN for steel.
Heat input formula	$HI = (V \times A \times 60) / (1000 \times \text{travel speed}) \rightarrow$ kJ/mm — controls cooling rate and grain size.
Preheat / interpass temperature	Minimum temperature before/during welding to slow cooling and prevent HAZ cracking.

Week 15 — Weld Metallurgy & Failure

Heat-affected zone (HAZ)	Base metal heated above ~730°C but not melted — the most failure-prone weld region.
Dilution	The proportion of melted base metal mixed into the deposited weld metal.
Hydrogen-induced cracking (HIC) / cold cracking	Delayed cracking requiring three simultaneous conditions: susceptible microstructure, hydrogen, tensile stress.
Ductile fracture (microvoid coalescence, cup-and-cone)	Fracture with plastic deformation, dimpled surface.
Brittle fracture (cleavage, chevron marks, river marks)	Sudden fracture with no plastic deformation; flat, crystalline surface.

Week 16 — Corrosion

Four-element corrosion cell (anode, cathode, electrolyte, metallic path)	All four must be present for electrochemical corrosion — remove one to stop it.
Galvanic series	Ranking of metals by electrochemical potential — determines which metal corrodes when dissimilar metals contact.
Passivation	Formation of a stable, adherent oxide layer that slows corrosion (stainless, aluminum, titanium).
Pitting corrosion / PREN	Localized passive-film breakdown; Pitting Resistance Equivalent Number quantifies alloy resistance.
Cathodic protection (sacrificial anode vs. impressed current)	Makes the structure the cathode so corrosion occurs only at a sacrificial anode.

Course Wrap-Up — Semester Synthesis

Metallic bond	Electrons form a shared 'electron sea' — explains conductivity and ductility.
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